Black rot on Apple

Botryosphaeria obutusa Black rot fungi



Shrivelled infected fruit (Bruce Watt, University of Maine/via Bugwood.org - CC BY-NC 3.0 US)



Black rot on apple fruit. (University of Georgia Plant Pathology/via Bugwood.org - CC BY 3.0 US)



Frogeye leaf spot (Circular spots with purplish or reddish edges and light tan interiors) on apple leaves affected by black rot. (Paul Bachi, University of Kentucky Research and Education Center/via Bugwood.org - CC BY 3.0 US)

Prevention	Monitoring	Direct Control	Direct Control	Restrictions
• Plant disease free apple seedlings from certified nursery operators such as KALRO and Wambugu farm in Olkalou • Establish apple orchards in well drained soils as waterlogging increases susceptibility to black rot • Remove dead wood or stumps of any apple trees cut down from the orchard as dead stumps can be a source of spores for future infections • Prune apple trees properly every year when during dormant period (see direct control) • Irrigate apple trees grown in sandy soils and for other soil types irrigate during dry spells • Clean farm tools thoroughly after use on infected plants using jik solution (3 to 5 ml jik in1 litre of water)	• Look for large brown rotten areas with firm and leathery flesh and concentric rings that alternate between black and brown anywhere on the fruit • Check for shriveled and dried out fruits that remain attached on the tree • Look for fruits that ripen early and are rotten at the core • Look for circular spots with purplish or reddish edges and light tan interiors on the leaves • Check for branch dieback and cankers which appear as sunken,reddish - brown areas on infected branches • Consider direct control actions when first symptoms are seen any leaf	• Prune out dead or infected branches, dried and shriveled fruits that remain on trees and destroy by burning • Destroy all infected plant materials by burning	• When using a pesticide, always wear protective clothing and follow the instructions on the product label	
			• Do not use chemicals with the same mode of action year after year as this can lead to resistance	
			• Always consult the most recent list of registered pesticides (PCPB)	
			• Spray Mancozeb such as BIOTHANE or AMICOP 50 WP or MILTHANE SUPER at the rate of 50 ml/20 L of water.	• WHO Class U (unlikely to cause acute harzard). PHI: 14 days, REI: 24 days. Repeat every 15 days.
			• Spray propineb 70 m/m such as ANTRACOL WP 70 at the rate of 50g/20L	• WHO class III (Slightly hazadous). PHI 14 days. Repeat at 7 to 14 days intervals
			• Spray Carbendazim 500g/L such as BENDAZIM 500 SC or RODAZIM SC at the rate of 20 ml/20L water	• WHO class III (Slightly hazadous). PHI 14 days. Repeat at 5 to 7 days intervals
			• Spray copper oxychloride 500g/kg e.g AMICOP 50 WP or COBOX 50 WP or GREEN COP 500WP at the rate of 40 - 60g/20L water	• WHO Class II Moderately hazardous; REI: 24hours
			• Spray copper based fungicides (Mancozeb 120g/kg + Cymoxanil 42g/kg + copper oxychloride) e.g TRINITY GOLD 425 WP or Cymoxanil 42g/kg + copper oxychloride) e.g COLONIZER 440 WP at the rate of 40 - 60g/20L water	• WHO Class II Moderately hazardous; REI: 24hours

Kenya

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